Phys 115A

Zih-Yu Hsieh

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Question 1. ***Clever Commutation***
Suppose I give you a quantum system that can be described with a raising operator a^+ and a lowering operator a^- that satisfy the commutation relation $[a^-, a^+] = 1$ (it doesn't have to be the quantum harmonic oscillator, specifically). You know that there is a properly- normalized state ψ that is annihilated by a^-, i.e., you know that $a^-\psi = 0$, but I don't tell you how a^+ acts on ψ.
(a) I prepare for you a state

$$\Psi = c \times (a^-a^-a^-a^-a^-a^+a^+a^+a^+\psi)$$

where c is a real constant. Is Ψ a normalizable state? If so, find c such that Ψ is properly normalized. What is the relation between Ψ and ψ?
(b) I prepare for you a different state

$$\Psi = c \times (a^-a^-a^-a^-a^+a^+\psi)$$

where c is a real constant. Is this new Ψ a normalizable state? If so, find c such that Ψ is properly normalized.

Proof.

(a) Given that $[a^-, a^+] = 1$ (or identity operator), we have $a^-a^+ = 1 + a^+a^-$, so the relation $a^-a^+\psi = \psi + a^+a^-\psi = \psi$, hence ψ is an eigenvector of the linear operator a^-a^+ with eigenvalue 1. Which, if consider the vector $a^+\psi$, we get the following:

$$a^-a^+(a^+\psi) = a^+\psi + a^+(a^-a^+\psi) = a^+\psi + a^+\psi = 2(a^+\psi) \tag{1}$$

Hence, $a^+\psi$ is an eigenvector of a^-a^+ with eigenvalue 2. Inductively, suppose given $n \in \mathbb{N}$ one has $(a^+)^n\psi$ being an eigenvector of a^-a^+ with eigenvalue $(n+1)$, then for the case of $(n+1)$, the following is true:

$$\begin{aligned} a^-a^+((a^+)^{n+1}\psi) &= (a^+)^{n+1}\psi + a^+(a^-(a^+)^{n+1}\psi) = (a^+)^{n+1}\psi + a^+((a^-a^+)(a^+)^n\psi) \quad (2) \\ &= (a^+)^{n+1}\psi + (n+1)a^+(a^+)^n\psi = (n+2)(a^+)^{n+1}\psi \quad (3) \end{aligned}$$

Hence, by induction, all $n \in \mathbb{N}$ satisfies $(a^+)^n\psi$ being an eigenvector of a^-a^+ with eigenvalue $(n+1)$.

With some simplification, another round of induction can be done on the vector of the form $\Psi = (a^-)^n(a^+)^n\psi$: For $n = 2$, one has $(a^-)(a^-a^+)(a^+\psi) = 2a^-a^+\psi = 2\psi$. Which, inductively suppose given $n \in \mathbb{N}$ one has $(a^-)^n(a^+)^n\psi = n!\psi$, then for the case of $(n+1)$, the following is true:

$$(a^-)^{n+1}(a^+)^{n+1}\psi = (a^-)^n(a^-a^+)(a^+)^n\psi = (n+1)(a^-)^n(a^+)^n\psi = (n+1)!\psi \tag{4}$$

Hence, by induction, all $n \in \mathbb{N}$ satisfies $(a^-)^n(a^+)^n\psi = n!\psi$. So, with $\Psi = c(a^-)^4(a^+)^4\psi$, one has $\Psi = c \cdot 4!\psi$, hence with $c! = 0$, Ψ (viewed as a state) is normalizable. With ψ being normalized, it requires $c = \frac{1}{4!}$ for Ψ to be properly normalized.

(b) Adapt the machinery built in part (a), given $\Psi = c(a^-)^4(a^+)^2\psi = c(a^-)^2((a^-)^2(a^+)^2\psi)$, it reduces to $\Psi = c(a^-)^2(2!\psi) = 0$ (since $a^-\psi = 0$). Hence, Ψ is the zero-vector, which is not normalizable.

$\square$

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Question 2. ***Constructing States (modified Griffiths 2.10 and 2.14)***
Given the explicit solution for the ground state of the harmonic oscillator,

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$$

(a) Construct $\psi_2(x)$ explicitly using the algebra formalism. (Hint: $\psi_1(x)$ is constructed this way in Griffiths Example 2.4.)
(b) Check the orthogonality of ψ_0, ψ_1, and ψ_2 by explicit integration. Hint: If you exploit the evenness and oddness of the functions, there is really only one integral left to do.
(c) Find the classical turning points for ψ_0, ψ_1, and ψ_2 (i.e., the values of x where the kinetic energy goes to zero for a classical harmonic oscillator of known total energy).
(d) Sketch ψ_0, ψ_1, ψ_2, ψ_3. For each state, mark the classical turning point. How does the classical turning point relate to the form of the wavefunction?
(e) For ψ_0, what is the probability (correct to three significant digits) of finding the particle outside the classically allowed region, meaning outside the classical turning points? You can use a math table under "Normal Distribution" or Mathematica or any other tool you wish to numerically evaluate the relevant integral.

Proof. For this question, recall the definition of the operators: $a_\pm = \frac{1}{\sqrt{2\hbar mw}}(mw\hat{x} \mp i\hat{p})$, $\hat{x}$ is the operator simply by multiplying x, and $\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x}$. So, $a_\pm = \frac{1}{\sqrt{2\hbar mw}}(mwx \mp \hbar\frac{\partial}{\partial x})$.

(a) Recall that $\psi_n(x) = \frac{1}{\sqrt{n!}}(a_+)^n\psi_0(x)$. So, for $n = 2$, one derives the following:

$$a_+\psi_0(x) = \frac{1}{\sqrt{2\hbar mw}}\left(mwx - \hbar\frac{\partial}{\partial x}\right)\left(\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}\right) \tag{5}$$

$$= \sqrt{\frac{mw}{2\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2} + \sqrt{\frac{\hbar}{2mw}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}}\frac{mw}{\hbar}xe^{-\frac{mw}{2\hbar}x^2} \tag{6}$$

$$= 2\sqrt{\frac{mw}{2\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2} = \sqrt{\frac{2mw}{\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2} \tag{7}$$

$$(a_+)^2\psi_0(x) = \frac{1}{\sqrt{2\hbar mw}}\left(mwx - \hbar\frac{\partial}{\partial x}\right)\left(\sqrt{\frac{2mw}{\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2}\right) \tag{8}$$

$$= \frac{mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2e^{-\frac{mw}{2\hbar}x^2} - \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2} + \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} \cdot \frac{mw}{\hbar}x^2e^{-\frac{mw}{2\hbar}x^2} \tag{9}$$

$$= \frac{2mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2e^{-\frac{mw}{2\hbar}x^2} - \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2} \tag{10}$$

So, we get that $\psi_2(x) = \frac{1}{\sqrt{2}}(a_+)^2\psi_0(x) = \frac{\sqrt{2}mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2e^{-\frac{mw}{2\hbar}x^2} - \frac{1}{\sqrt{2}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}$.

(b) It suffices to check orthogonality for $\psi_0, (a_+)\psi_0, (a_+)^2\psi_0$ (since scaling doesn't affect orthogonality).

For $\psi_0, (a_+)\psi_0$, the integration provides the following:

$$\int_{-\infty}^{\infty}((a_+)\psi_0(x))^*\psi_0(x)dx = \int_{-\infty}^{\infty}\sqrt{\frac{2mw}{\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2}\cdot\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}dx = 0 \tag{11}$$

(Note: The above is an integration of an odd function that converges, hence provides 0). Hence, $\psi_0, (a_+)\psi_0$ are orthogonal under integration of inner product.

For $(a_+)\psi_0, (a_+)^2\psi_0$, the integration provides the following:

$$\int_{-\infty}^{\infty}((a_+)\psi_0(x))^*((a_+)^2\psi_0(x))dx \tag{12}$$

$$= \int_{-\infty}^{\infty}\sqrt{\frac{2mw}{\hbar}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2}\cdot\left(\frac{\sqrt{2}mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2e^{-\frac{mw}{2\hbar}x^2} - \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}\right)dx \tag{13}$$

$$= K\int_{-\infty}^{\infty}x^3e^{-\frac{mw}{\hbar}x^2}dx + L\int_{-\infty}^{\infty}xe^{-\frac{mw}{\hbar}x^2}dx = 0 \tag{14}$$

(Note: Above are two odd function that has converging integrals, which with domain being symmetric it converges to 0. K, L are proper substitutions of the constants). Hence, $(a_+)\psi_0, (a_+)^2\psi_0$ are also orthogonal under integration inner product.

Finally, for $\psi_0, (a_+)^2\psi_0$, the integration provides the following:

$$\int_{-\infty}^{\infty}(\psi_0(x))^*((a_+)^2\psi_0(x))dx \tag{15}$$

$$= \int_{-\infty}^{\infty}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}\left(\frac{\sqrt{2}mw}{\hbar}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} x^2e^{-\frac{mw}{2\hbar}x^2} - \frac{1}{\sqrt{2}}\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}\right)dx \tag{16}$$

$$= -\frac{1}{\sqrt{2}}\int_{-\infty}^{\infty}\sqrt{\frac{mw}{\pi\hbar}}e^{-(\sqrt{\frac{mw}{\hbar}}x)^2}dx + \frac{\sqrt{2}mw}{\hbar}\int_{-\infty}^{\infty}\sqrt{\frac{mw}{\pi\hbar}}x^2e^{-\frac{mw}{\hbar}x^2}dx \tag{17}$$

$$= -\frac{1}{\sqrt{2}} - \frac{\sqrt{2}mw}{\hbar}\left(\int_{-\infty}^{\infty}\sqrt{\frac{mw}{\pi\hbar}}\frac{\partial}{\partial\lambda}\left(e^{-\lambda x^2}\right)dx\right)\Bigg|_{\lambda=\frac{mw}{\hbar}} \tag{18}$$

$$= -\frac{\sqrt{2}mw}{\hbar}\frac{d}{d\lambda}\Bigg|_{\lambda=\frac{mw}{\hbar}}\int_{-\infty}^{\infty}\sqrt{\frac{mw}{\pi\hbar}}e^{-\lambda x^2}dx - \frac{1}{\sqrt{2}} = -\frac{\sqrt{2}mw}{\hbar}\sqrt{\frac{mw}{\pi\hbar}}\sqrt{\frac{\pi}{\lambda}}\Bigg|_{\lambda=\frac{mw}{\hbar}} - \frac{1}{\sqrt{2}} \tag{19}$$

$$= -\sqrt{\frac{2}{\pi}}\left(\frac{mw}{\hbar}\right)^{\frac{3}{2}}\cdot\left(-\frac{1}{2}\frac{\sqrt{\pi}}{\lambda^{\frac{3}{2}}}\right)\Bigg|_{\lambda=\frac{mw}{\hbar}} - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0 \tag{20}$$

Hence, under integration inner product, $\psi_0, (a_+)^2\psi_0$ are also orthogonal.

Since ψ_0, ψ_1, ψ_2 are scalar multiples of $\psi_0, (a_+)\psi_0, (a_+)^2\psi_0$ respectively, hence they're also mutually othogonal under integration inner product.

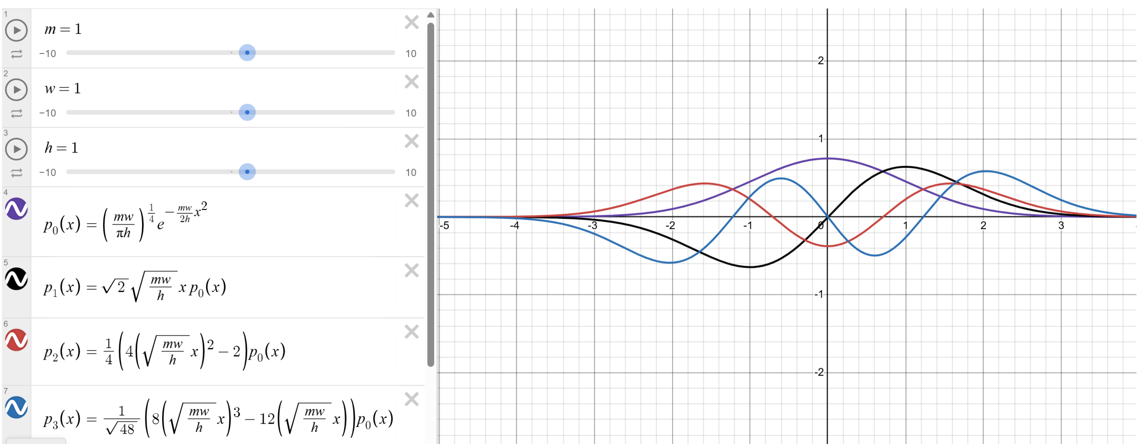
(c) Given the energy of $\psi_n(x)$ is $E_n = \left(n + \frac{1}{2}\right)\hbar w$, and the potential energy is $V(x) = \frac{1}{2}mw^2x^2$, the classical turning points occurs when $E_n = V(x)$, providing the following solution:

$$\frac{1}{2}mw^2x^2 = \left(n + \frac{1}{2}\right)\hbar w \implies x = \pm\sqrt{\left(n + \frac{1}{2}\right)\frac{2\hbar}{mw}} = \pm\sqrt{\frac{(2n+1)\hbar}{mw}} \tag{21}$$

Hence, the classical turning points of the four states ψ_0, ψ_1, ψ_2 (denoted as $\pm x_0, \pm x_1, \pm x_2$ respectively) are given as:

$$\pm x_0 = \pm\sqrt{\frac{\hbar}{mw}}, \quad \pm x_1 = \pm\sqrt{\frac{3\hbar}{mw}}, \quad \pm x_2 = \pm\sqrt{\frac{5\hbar}{mw}} \tag{22}$$

(d) Here are some graphs of the first four states (starting from ground states):



Here, $\psi_0(x)$ is in purple, $\psi_1(x)$ is in black, $\psi_2(x)$ is in red, and $\psi_3(x)$ is in blue.

Based on calculation from the previous part, graphically it is verified that for all states (besides psi_0) above, the classical turning points correspond to the farthest local extrema of the functions (which are the local min / max that are the farthest away from the origin). As interpretation, since $|\psi|^2$ represents the probability of appearing at the point as position, and classically the turning points are where the particles are "most probable" to appear, which these farthest extremas are also global extrema, hence corresponds to the most probable position for the particle to appear.

For ψ_0 instead, the classical turning point instead reflects the variance - which are the uncertainties of the particle's position away from the origina (which is the mean, or the average position of the particle).

(e)

$\square$

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Question 3. ***3 Uncertainty Principle for the nth state (modified Griffiths 2.12)***
Consider the nth stationary state of the harmonic oscillator.
(a) Find $\langle x\rangle$, $\langle p\rangle$, $\langle x^2\rangle$ and $\langle p^2\rangle$ for this state, following the method of Example 2.5.
(b) Find the expectation value of the kinetic energy, $\langle T\rangle$, and the expectation value of the potential energy, $\langle V\rangle$, for this state.
(c) Compute $\sigma_x\sigma_p$ for this state, and check that the uncertainty principle is satisfied. Under what circumstances is it saturated (i.e., is $\sigma_x\sigma_p = \hbar/2$)?
Hint: For this problem, do not explicitly integrate the wavefunction, which has a terrible form for arbitrary n. Instead, notice that you can write x and p in terms of raising and lowering operators,

$$x = \sqrt{\frac{\hbar}{2m\omega}}(a^+ + a^-), \quad p = i\sqrt{\frac{\hbar m\omega}{2}}(a^+ - a^-).$$

Then to compute the expectation values you just have to act on ψ_n with raising and lowering operators, which you know how to do.

Proof. □

Question 4. ***4 Back of the envelope***
Using the symmetry of the potential, come up with an expression for the zero-point energy of a harmonic oscillator on the assumption that it is the minimum energy required by the uncertainty principle.

Proof. $\square$

Question 5. ***5 Coherent states***

In problem 3, you found that most of the stationary states of the harmonic oscillator satisfy, but do not saturate, the uncertainty principle. Now let us consider a special superposition of these stationary states called a coherent state, which has some fairly interesting properties. We call such a state an eigenfunction of the lowering operator. It just means the coherent state $S_\alpha(x)$ satisfies $a^- S_\alpha(x) = \alpha S_\alpha(x)$ where α is a complex number. Note that there are no normalizable eigenfunctions of the raising operator.

(a) Calculate $\langle x \rangle$ and $\langle p \rangle$ for the state $S_\alpha(x)$. Does this remind you of anything from previous problems in this set? Hint: Since you know how a^- acts on $S_\alpha(x)$, use the fact that

$$\int_{-\infty}^{\infty} f^*(x)(a^{\pm} g(x))\,dx = \int_{-\infty}^{\infty} (a^{\mp} f(x))^* g(x)\,dx.$$

Also, keep in mind that α is in general complex.

(b) Calculate $\langle x^2 \rangle$ and $\langle p^2 \rangle$ for the state $S_\alpha(x)$.

(c) Show that S_α saturates the uncertainty principle.

(d) We can express $S_\alpha(x)$ at $t = 0$ explicitly in terms of the stationary states ψ_n,

$$S_\alpha(x) = \sum_{n=0}^{\infty} c_n \psi_n(x)$$

Show that $c_n = \frac{\alpha^n}{\sqrt{n!}} c_0$. Hint: Keep using the earlier hint!

(e) Fix c_0 by normalizing $S_\alpha(x)$.

(f) Now consider $S_\alpha(x,t)$,

$$S_\alpha(x,t) = \sum_{n=0}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}.$$

Show that $S_\alpha(x,t)$ remains an eigenfunction of a^- but that α now evolves in time. That is, show that $a^- S_\alpha(x,t) = \alpha(t) S_\alpha(x,t)$, where $\alpha(t) = e^{-i\omega t}\alpha$. Argue that this means coherent states stay coherent in time, and continue to saturate the uncertainty principle.

(g) Can we think of the ground state ψ_0 as a coherent state? What is α in this case?

These states are profoundly interesting in quantum optics, since they provide an accurate quantum description of laser light. Roy Glauber was awarded half of the Nobel Prize in Physics in 2005 for his 1963 work on understanding the properties of coherent states in quantum optics.

Proof. □